Exp: 8

Simple chatting program in computer networks

Aim: To develop a simple chatting application using python socket Programming

Observation;

chatting allows users to exchange messages through a comp network socket programming provides communication blw two computers

The server waits for client connections, while the client converts to the serve. TCP ensures reliable data transmission

Procedure;

1. Install python & open VS code
2. create two files: server.py & client.py
3. import socket & threading Modules
4. create a TCP socket in both programs
5. Bind & start server, connect client
6. Run Server First & client next
7. exchange messages b/w two terminals
8. Type exit to end

code:

→Server

```
import socket
import threading
Server = socket.socket()
Server.bind(("127.0.0.1", 12345))
Server.listen(1)
Print("waiting for client...")
def receive():
    while True:
        msg = conn.req(1024).decode()
        if not msg or msg
            break
        Print("client:", msg)
threading.Thread(target=receive,
                 start())
while True:
    msg = input("server:")
    conn.sendall(msg.encode())
    if msg.lower() == "exit":
        break
Conn.close()
server.close()
```

client

```
import socket
import threading
client=socket.socket()
client=connect(("127.0.0.1", 12345))
Print("connected to server")

def receive():
    while True:
        msg=client.rec(1024).decode(
        if not msg or msg.lower()=="exi
            break
        Print("server:", msg)
threading.Thread(target=receive,
                 start())
while True:
    msg=input("client:")
    client.endall(msg.encode())
    if msg.lower()=="exit":
        break
client.close()
```

Server:

```
waiting for client
client connected
Server: Hello
client: Hi
Server: How are you?
Client: I am fine
```

client:
connected to server
client: Hi
Server: Hello
client: I am fine
Server: How are you?

Result: Thus, a simple chatting application was implemented successfully using python socket programming & TCP to exchange messages between client & server.